

There are a few things I should clarify or maybe remind:

1. When using the initialization file in the “input file” folder, that is #6 stated below, such would generate a txt file in the input file folder. And then you should use the first code base (AGVertex Model) to relax it first. And then use the last timestep of that txt file as the input file for relaxation(Vertexmodelreid or Vertexmodel by Ben Deacon). If you tried to run a txt file via visual studio and it just simply quit, it is possible that your path is not correct.
2. The functions would only work if the file is in the folder where the function is. Note that there might be some mismatching that certain txt file only run when it is copied and pasted into a specific folder. If such a thing occurs, you could check the matlab error message to figure out where you should put the txt file and run it.

In input file folder

Vertex Model Analysis Toolkit — Summer Handoff Notes

Overview

This repository contains MATLAB scripts for analyzing and visualizing epithelial vertex model simulations. The tools focus on:

- * Tissue geometry visualization
- * Edge tension analysis
- * Greyplot generation
- * Nucleation statistics
- * Movie generation
- * Final timestep extraction
- * Image merging for publication figures

The simulation outputs are `.txt` files generated by the C++ vertex model code.

1. Extract Second-to-Last Timestep

Script

``ExtractSecondLastTimestep.m``

Purpose

Extracts the second-to-last complete timestep from a very large simulation output file.

This is useful because:

- * simulations may terminate during writing,
- * the final timestep can occasionally be incomplete/corrupted,
- * many analyses should use the last fully written configuration.

Example Input

```
``matlab
inputFile = '12_out.txt';
outputFile = '12_lasttimestep.txt';
``
```

Output

A new text file containing only the second-to-last timestep.

2. Moving Greyplot Movie

Script

``MovingGreyPlot.m``

Purpose

Creates:

- * a movie,
- * and saved image frames

showing how the distribution of vertex geometry evolves over time.

The greyplot compares:

- * **x-axis:** bisector orientation (``psi``)
- * **y-axis:** opening angle (``theta``)

Scientific Meaning

Each pixel corresponds to a geometric state of a tricellular junction (TCJ).

Bright regions mean:

> many TCJs exist with that geometry.

Dark regions mean:

> few or no TCJs exist with that geometry.

This helps identify:

- * preferred geometries,
- * anisotropy,
- * alignment,
- * emergent structure during tissue evolution.

3. greyplothistogram

Script:

greyplothistogram.m

The important idea is:

every tricellular junction (TCJ) contributes a point:

(ψ, θ)

where

ψ = bisector orientation

θ = opening angle

The script then bins those points into a 2D grid.

What the Greyplot Actually Is

Mathematically: $H_C = \text{histcounts2}(\text{psi_all}, \text{theta_all}, \text{psi_edges}, \text{theta_edges})$;

Means: Count how many TCJs fall into each small rectangle in (ψ, θ) space.

Then: $H_C = H_C / \text{sum}(H_C(:))$; normalizes it into a probability distribution.

So each pixel becomes: $P(\psi, \theta)$

Bright Pixel Means:

many TCJs have this geometry.

Dark Pixel Means: few TCJs have this geometry.

Panel-by-Panel Meaning

Panel C — All TCJs

H_C

This is: $P(\psi, \theta)$, the raw geometric distribution.

It answers: “What kinds of tricellular junctions commonly exist in the tissue?”

Nucleation condition

A vertex nucleates if: at least two edge tensions exceed the threshold T^*

The nucleating angle is defined as:

the angle between the two high-tension edges

Panel D — Nucleation Events

H_D

This only includes: vertices where tensions exceed threshold, and specifically the nucleating angle.

This answers: “What geometries are associated with nucleation?”

Bright spot means: many nucleation events happen at this geometry.

Panel E — Relative Nucleation Rate

$H_E = H_D / H_C$;

This is the important physics panel.

It asks: “Given that a TCJ has geometry (ψ, θ) , how likely is it to nucleate?”

That identifies:

mechanically unstable configurations,

stress-focusing geometries,

preferred failure states.

Physically What You Might Observe

If opening angle matters

You may see:

bright horizontal bands.

Meaning: nucleation strongly depends on opening angle.

If orientation matters

You may see: vertical structure.

Meaning: tissue anisotropy or alignment effects.

If both matter

You may see: isolated hotspots.

Meaning: specific geometries are especially unstable.

Purpose

greyplothistogram.m analyzes tissue simulation output files and generates three greyplot panels that characterize the geometry of tricellular junctions (TCJs) and where nucleation events occur.

The script aggregates data across multiple simulations (different random seeds or packings).

It produces:

Panel C — Distribution of all TCJ geometries

Panel D — Distribution of nucleating TCJs

Panel E — Relative nucleation probability

These plots help identify whether certain TCJ geometries are more likely to nucleate high-tension events.

Physical Meaning

Each tricellular junction (TCJ) has:

Opening angle θ

Internal angle between two edges

Range: $0 \rightarrow \pi$

Bisector orientation ψ

Direction the angle points

Range: $0 \rightarrow \pi/2$

Usage

Step 1 — Place simulation outputs

Put all simulation output files in the same folder.

Example:

1_out.txt

3_out.txt

10_out.txt

11_out.txt

12_out.txt

Step 2 — Edit file list

Modify:

filelist = {

'1_out.txt'

'3_out.txt'

'10_out.txt'

};

to include the simulations you want analyzed.

Step 3 — Set nucleation threshold

Threshold is defined by:
Y_param = 5.66;
EpsilonMax = 0.6;
T_star = 0.97 * Y_param * EpsilonMax;
A nucleation event occurs when:
two edge tensions > T_star
Changing this value changes the definition of nucleation.

Step 4 — Run script

Run:
greyplothistogram
The script will generate:
Panel C
Panel D
Panel E
Geometry conventions

The script uses the same conventions as the C++ simulation code:

Quantity	Meaning
theta	Opening angle
psi	Bisector orientation

Plot axes:
Axis Quantity
x-axis ψ
y-axis θ
Typical Observations
Bright region in Panel C Means: that geometry is common in the tissue
Bright region in Panel D Means: many nucleation events occur there

but this may simply reflect that the geometry is common.

Bright region in Panel E

Means: nucleation is enhanced for that geometry, geometry may mechanically favor instability

This is the key quantity for identifying preferred nucleation configurations.

Applications

This script is useful for:

studying tension localization
identifying mechanically unstable TCJs
comparing different simulation conditions
comparing different packing geometries
analyzing anisotropy in epithelial tissues
investigating geometry–mechanics coupling
Dependencies

The script assumes simulation outputs contain:

Vertices
Edges
Cells
CURRENTTIME

with vertex columns:

Angletheta1
Angletheta2
Angletheta3
PsiAngle1
PsiAngle2
PsiAngle3

and edge tensions stored in column 6 of the edge block.

4. Merge Images into Publication Figure

Script

`MergeImages.m`

Purpose

Combines multiple PNG images into a single tightly packed figure.

Useful for:

- * paper figures,
- * presentations,
- * comparing packings.

Input

Folder:

```matlab  
merge\_images  
```

Output

```matlab  
merged\_packings.png  
```

Features

- * automatic numerical sorting,
- * zero spacing between images,
- * export at high resolution.

5. Tension Histogram Evolution Movie

Script

`TensionHistogramMovie.m`

Purpose

Creates:

- * a movie,
- * and saved frames

showing how the edge tension distribution evolves over time.

Scientific Meaning

Tracks:

- * tension buildup,
- * stress redistribution,
- * emergence of high-tension populations,
- * relaxation dynamics.

Useful for identifying:

- * mechanical transitions,
- * nucleation onset,
- * tension broadening.

Output

* `TensionEvolution.mp4`
* PNG frames in:

``matlab
tension_frames
``

6. Initialization file

newlyupdated_initialization_file.m
Purpose

This script generates an initial tissue configuration for the vertex model simulation using a periodic Voronoi tessellation.

It creates:

vertices
edges
cells
periodic boundary flags
simulation parameters

and exports everything into a .txt initialization file that can be directly loaded into the C++ vertex model simulation.

The script is designed to produce mechanically stable initial packings with controlled geometry and reproducible random seeds.

In MatlabVis folder

6. CreateMovie.m and CreateMovie_m.m

Inside the createMovie, I have indicated two functions,
Plotsimulationoutput.m and plotsimulationoutput_modified.m

Plotsimulationoutput.m: (IN CreateMovie.m)
Purpose: Show the actual tension value of every edge.
Uses a continuous color scale (hot colormap).
Low-tension edges appear dark.
High-tension edges appear bright.
Includes a colorbar showing the numerical tension range.
Useful when you want to see the full distribution of tensions throughout the tissue.

Plotsimulationoutput_modified.m (In CreateMovie_m.m)
Purpose: Show whether an edge is above or below a nucleation threshold.
Uses only two colors.
Edges with tension greater than the threshold are one color (e.g. red).
Edges with tension less than or equal to the threshold are another color (e.g. blue).
No continuous tension information is shown.
Makes it easy to identify which edges are considered "high-tension" according to the nucleation criterion.

Purpose

Creates movies of the tissue itself over time.

The script:

* reads simulation output,
* reconstructs periodic boundaries,
* plots edges,
* exports TIFF frames.

Main Visualization

Edges are colored according to tension.

Important Options

```
```matlab
CreateMovie(filename, tension_bounds)
```
```

tension_bounds = 1

Uses global min/max tension across the full simulation.

tension_bounds = 0

Uses only the final timestep tension range.
